

Qualifying Exam in Microeconomics
Arizona State University
June 2018

There are four questions and you have four hours to complete the exam. If you believe that a question is in error, please explain the error and continue working.

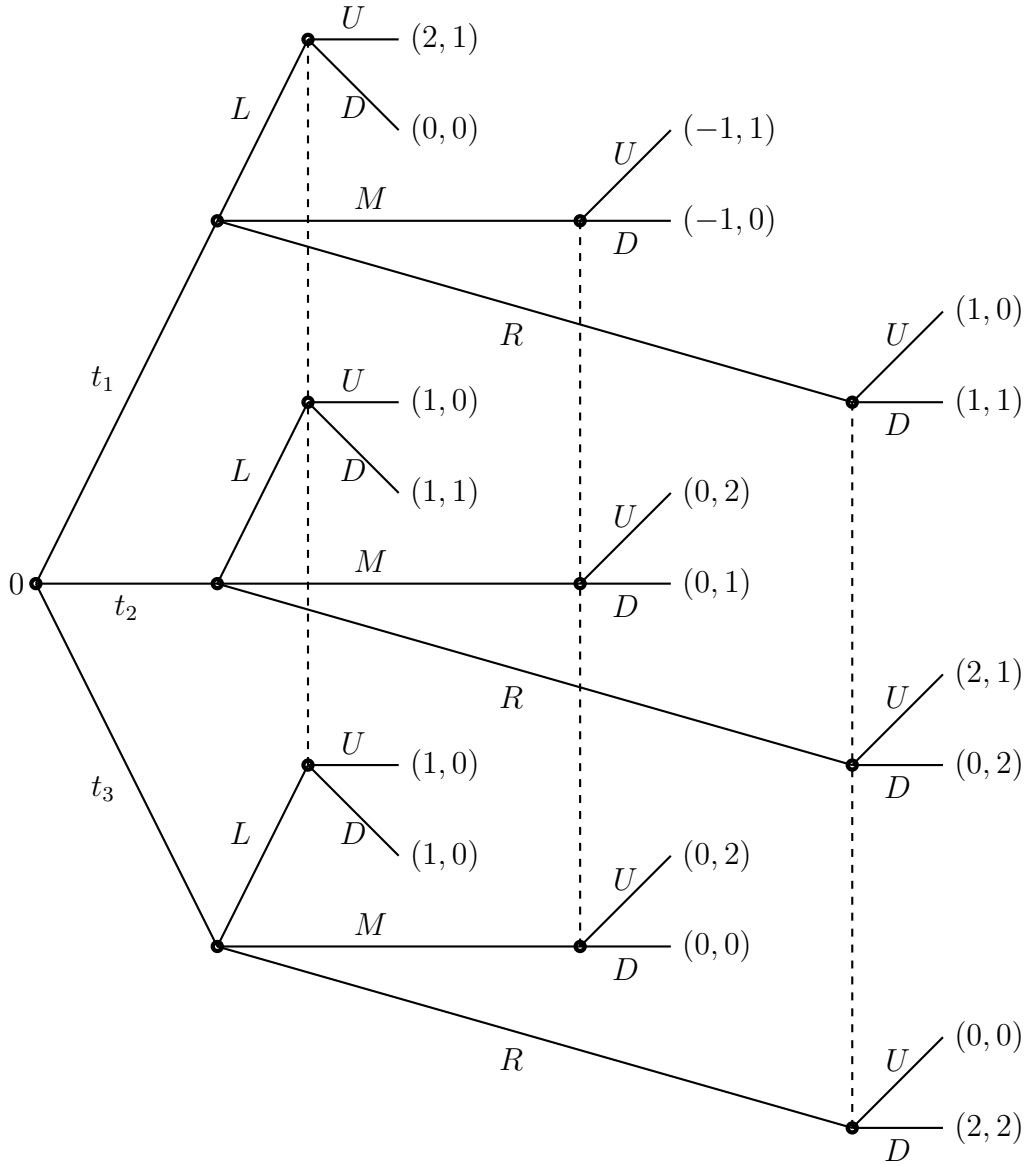
Questions 1 and 2 consider a consumer with preferences on the consumption set $X = \mathbb{R}_+^2$ represented by a function $u(x_1, x_2) = f(x_1) + x_2$, where $f : \mathbb{R}_+ \rightarrow \mathbb{R}$ is continuous. The price vector is $p = (p_1, p_2) \gg 0$ and the consumer's wealth is $w > 0$.

1. Answer the following three questions about this consumer.

- (a) Prove that u is quasiconcave *if and only if* f is concave.
- (b) Suppose that $f(x_1) = -e^{-x_1}$ for $x_1 \geq 0$. Find the associated Walrasian (ordinary) demand function. (Remember that the consumption set is $X = \mathbb{R}_+^2$.)
- (c) Consider an exchange economy with a single consumer with preferences from part (b) and endowment $(\omega_1, \omega_2) = (0, 2)$.
 - i. Formally define the notion of a competitive equilibrium for *this* economy.
 - ii. Identify *every* competitive equilibrium for this economy. Is the competitive equilibrium relative price (p_1/p_2) unique?

2. Return to the case of $u(x_1, x_2) = f(x_1) + x_2$ for a general continuous function f . Prove that good 1 is *not a Giffen good* for this consumer in this sense: fixing $(p_2, w) \gg 0$, if x^t solves the consumer's problem at $p_1 = p_1^t$ for $t = 0, 1$ and $0 < p_1^1 < p_1^0$, then $x_1^1 \geq x_1^0$; in words, if the price of good 1 *decreases*, then the demand for good 1 *does not decrease*.

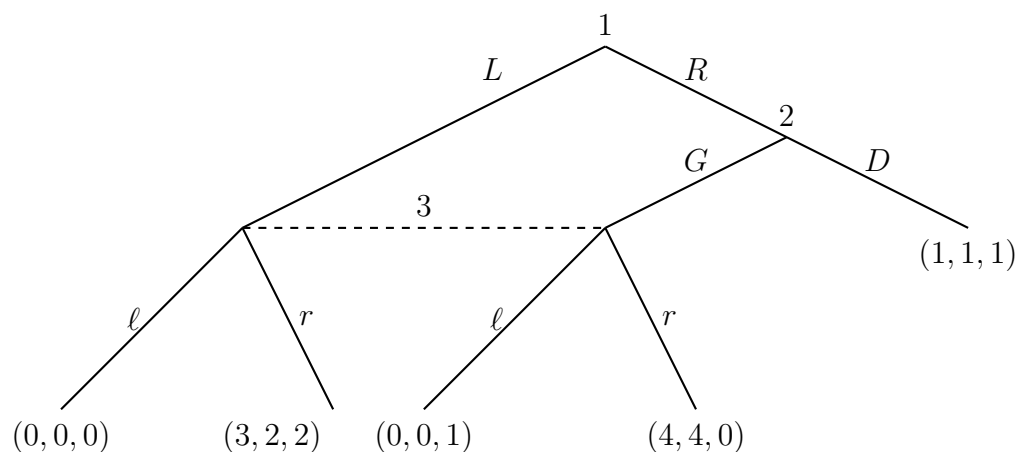
3. The game below is a standard signaling game. Nature chooses with equal probability player 1's type from the set $T = \{t_1, t_2, t_3\}$. Player 1's type is the player's private information. After observing $t_i \in T$, player 1 chooses an action from the set $A = \{L, M, R\}$. After observing player 1's chosen action, player 2 chooses U or D . Payoffs are indicated as ordered pairs, the first number is player 1's payoff.



- (a) Define each player's set of pure strategies and state how many pure strategies are available to each player. Please use s to denote a generic pure strategy for player 1 and r to denote a generic pure strategy for player 2; use S_1 and S_2 to denote the respective sets of pure strategies.

- (b) Define pure-strategy Weakly Perfect Bayesian Equilibrium (WPBE) *in this game*.
- (c) Find all pure-strategy WPBE.
- (d) Is there a separating pure-strategy WPBE? And a pooling pure-strategy WPBE?

4. Consider the following three-person game.



- (a) Find *all* (i.e. pure and mixed strategy) Nash equilibria (NE) of this game. Please denote by α the probability that player 1 chooses L , by β the probability that player 2 chooses G , and by γ the probability that player 3 chooses ℓ .
- (b) Define Sequential Equilibrium (SE) for this game.
- (c) Is there any SE in pure or mixed strategies where player 1 plays L with probability $\alpha \in (0, 1]$?